27.5 mm

27.5 mm

137313

IFIMOL* IV

Paracetamol Solution for Intravenous Infusion
10 mg/mL

COMPOSITION
Each mL contains:
Paracetamol BP……10mg
Water for Injection BP…..q.s.

DESCRIPTION
A clear colorless sterile solution.

PHARMACODYNAMICS
Pharmacotherapeutic group: OTHER ANALGESICS AND ANTIPYRETICS, ATC code: N02BE01
The precise mechanism of the analgesic and antipyretic properties of paracetamol has yet to be established; it may involve central and peripheral actions.
IFIMOL IV provides onset of pain relief within 5 to 10 minutes after the start of administration. The peak analgesic effect is obtained in 1 hour and the duration of this effect is usually 4 to 6 hours.
IFIMOL IV reduces fever within 30 minutes after the start of administration with a duration of the antipyretic effect of at least 6 hours.

PHARMACOKINETICS
Adults:
Absorption:
Paracetamol pharmacokinetics is linear up to 2 g after single administration and after repeated administration during 24 hours.
The bioavailability of paracetamol following infusion of 500 mg and 1 g of IFIMOL IV is similar to that observed following infusion of 1 g and 2 g propacetamol (corresponding to 500 mg and 1 g paracetamol respectively). The maximal plasma concentration (Cmax) of paracetamol observed at the end of 15-minutes intravenous infusion of 500 mg and 1 g of IFIMOL IV is about 15 µg/mL and 30 µg/mL respectively.
Distribution:
The volume of distribution of paracetamol is approximately 1 L/kg.
Paracetamol is not extensively bound to plasma proteins.
Following infusion of 1 g paracetamol, significant concentrations of paracetamol (about 1.5 µg/mL) were observed in the Cerebro Spinal Fluid as and from the 20 th minutefollowing infusion.
Metabolism:
Paracetamol is metabolised mainly in the liver following two major hepatic pathways:
glucuronic acid conjugation and sulphuric acid conjugation. The latter route is rapidly saturable at doses that exceed the therapeutic doses. A small fraction (less than 4%) is metabolised by cytochrome P450 to a reactive intermediate (N-acetyl benzoquinone imine) which, under normal conditions of use, is rapidly detoxified by reduced glutathione and eliminated in the urine after conjugation with cysteine and mercapturic acid. However, during massive overdosing, the quantity of this toxic metabolite is increased.
Elimination:
The metabolites of paracetamol are mainly excreted in the urine. 90% of the dose administered is excreted in 24 hours, mainly as glucuronide (60-80%) and sulphate (20-30%) conjugates. Less than 5% is eliminated unchanged. Plasma half-life is 2.7 hours and total body clearance is 18 L/h.
Neonates, infants and children
The pharmacokinetic parameters of paracetamol observed in infants and children are similar to those observed in adults, except for the plasma half-life that is slightly shorter (1.5 to 2 h) than in adults. In neonates, the plasma half-life is longer than in infants i.e. around 3.5 hours. Neonates, infants and children up to 10 years excrete significantly less glucuronide and more sulphate conjugates than adults.
*Table. Age related pharmacokinetic values (standardized clearance, *CLstd/Foral($l.h^{-1}$ 70 kg^{-1}), are presented below.*

Age	*Weight (kg)*	*CL_{std}/F_{oral}($l.h^{-1}$ 70kg^{-1})*
40 weeks PCA	*3.3*	*5.9*
3 months PNA	*6*	*8.8*
6 months PNA	*7.5*	*11.1*
1 year PNA	*10*	*13.6*
2 years PNA	*12*	*15.6*
5 years PNA	*20*	*16.3*
8 years PNA	*25*	*16.3*

*CL_{std} is the population estimate for CL
Special populations:
Renal insufficiency
In cases of severe renal impairment (creatinine clearance 10-30 mL/min), the elimination of paracetamol is slightly delayed, the elimination half-life ranging from 2 to 5.3 hours.
For the glucuronide and sulphate conjugates, the elimination rate is 3 times slower in subjects with severe renal impairment than in healthy subjects. Therefore, it is recommended, when giving paracetamol to patients with severe renal impairment (creatinine clearance ≤ 30 mL/min), to increase the minimum interval between each administration to 6 hours (see section Recommended Dose).
Elderly subjects
The pharmacokinetics and the metabolism of paracetamol are not modified in elderly subjects. No dose adjustment is required in this population.

INDICATIONS
Paracetamol is indicated for the short-term treatment of moderate pain, especially following surgery, and for the short-term treatment of fever, when administration by intravenous route is clinically justified by an urgent need to treat pain or hyperthermia and/or when other routes of administration are not possible.

RECOMMENDED DOSE
Intravenous route.
Posology:
Dosing based on patient weight (please see the dosing table here below)

Patient weight	Dose per administration	Volume per administration	Maximum volume of IFIMOL IV (10 mg/mL) per administration based on upper weight limits of group (mL)***	Maximum Daily Dose **
⩽ **10 kg ***	7.5 mg/kg	0.75 mL/kg	7.5mL	30 mg/kg
> 10 kg to ⩽ 33kg	15 mg/kg	1.5mL/kg	49.5mL	60mg/kg not exceeding 2g
> 33 kg to ⩽ 50kg	15 mg/kg	1.5mL/kg	75 mL	60mg/kg not exceeding 3g
>50kg with additional risk factors for hepatotoxicity	1g	100mL	100mL	3g
> 50 kg and no additional risk factors for hepatotoxicity	1 g	100mL	100mL	4g

*** Pre-term newborn infants:** No safety and efficacy data are available for pre-term newborn infants (see section Pharmacokinetic).
** **Maximum daily dose:** The maximum daily dose as presented in the table above is for patients that are not receiving other paracetamol containing products and should be adjusted accordingly taking such products into account.
*** **Patients weighing less will require smaller volumes.**
The minimum interval between each administration must be at least 4 hours.
The minimum interval between each administration in patients with severe renal insufficiency must be at least 6 hours.
No more than 4 doses to be given in 24 hours.
Severe renal insufficiency:
It is recommended, when giving paracetamol to patients with severe renal impairment (creatinine clearance ≤ 30 mL/min), to increase the minimum interval between each administration to 6 hours (See section Pharmacokinetic).
In adults with hepatocellular insufficiency, chronic alcoholism, chronic malnutrition (low reserves of hepatic glutathione), dehydration:
The maximum daily dose must not exceed 3 g (see section Warnings and Precautions).
Method of administration:
Parenteral

> Take care when prescribing and administering IFIMOL IV to avoid dosing errors due to confusion between milligram (mg) and milliliter (mL), which could result in accidental overdose and death. Take care to ensure the proper dose is communicated and dispensed.
> When writing prescriptions, include both the total dose in mg and the total dose in volume. Take care to ensure the dose is measured and administered accurately.

1) Do not remove inner container from overwrap until ready for use.
2) The overwrap is a moisture barrier. The inner container maintains the sterility of the product.
3) After removing overwrap, check for minute leaks by squeezing the container. Do not use if leaks are found and return for replacement.
4) Bring the injection bottle to room temperature or preferably to 37^{0}C just before use.
5) Tare the top Aluminium protector of cap and clean the exposed surface with surgical spirit.
6) To keep their internal pressure in balance with the ambient pressure, insert a sterile injection cannula through the point marked on the rubber disc (indicated by a circle).
7) Insert the cannula of the sterile infusion set into the other hole in rubber disc.
8) For administration hang the bottle upside down.
9) The paracetamol solution should be administered as a 15minute intravenous infusion.

Caution: Even inviable damage to bottle caused during storage or transit, may result in contamination. Do not use if leak found on squeezing, or contents not clear, and return for replacement.

Incompatibilities:
IFIMOL IV should not be mixed with other medicinal products.
Special precautions for disposal and other handling:
Text for the 100ml bottle:
Use a 0.8 mm needle and vertically perforate the stopper at the spot specifically indicated.
Before administration, the product should be visually inspected for any particulate matter and discoloration. For single use only. Any unused solution should be discarded.
The diluted solution should be visually inspected and should not be used in presence of opalescence, visible particulate matters or precipitate.

CONTRAINDICATIONS
IFIMOL IV is contraindicated:
- in patients with hypersensitivity to paracetamol or to propacetamol hydrochloride (prodrug of paracetamol) or to one of the excipients.
- in cases of severe hepatocellular insufficiency.

WARNINGS & PRECAUTIONS
Warnings

> RISK OF MEDICATION ERRORS
> Take care to avoid dosing errors due to confusion between milligram (mg) and milliliter (mL), which could result in accidental overdose and death (see section Recommended Dose).

It is recommended that a suitable analgesic oral treatment be used as soon as this route of administration is possible.
In order to avoid the risk of overdose, check that other medicines administered do not contain either paracetamol or propacetamol.
Doses higher than those recommended entail the risk of very serious liver damage.
Clinical signs and symptoms of liver damage are not usually seen until two days, and up to a maximum of 4-6 days, after administration. Treatment with antidote should be given as soon as possible (see section Symptoms and Treatment of Overdose).

> Warning: This preparation contains PARACETAMOL.
> Do not take any other paracetamol containing medicine at the same time.
> Allergy alert: Paracetamol may cause severe skin reactions. Symptoms may include skin reddening, blisters or rash.
> These could be signs of a serious condition. If these reactions occur, stop use seek medical assistance right away.

> **WARNING**
> **This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients.**
> **Do not use if known to be hypersensitive to bisulphites.**

Precautions for use
Paracetamol should be used with caution in cases of:
- hepatocellular insufficiency,
- severe renal insufficiency (creatinine clearance <30mL/min),
- chronic alcoholism,
- chronic malnutrition (low reserves of hepatic glutathione),
- dehydration.

Effects on Ability to Drive and Use Machines
It is unlikely to impair a patient's ability to drive or use machinery.

INTERACTIONS WITH OTHER MEDICAMENTS
- Probenecid causes an almost 2-fold reduction in clearance of paracetamol by inhibiting its conjugation with glucuronic acid. A reduction of the paracetamol dose should be considered for concomitant treatment with probenecid,
- Salicylamide may prolong the elimination $t_{1/2}$ of paracetamol,
- Caution should be paid to the concomitant intake of enzyme-inducing substances (see section Symptoms and Treatment of Overdose).
- Concomitant use of paracetamol (4 g per day for at least 4 days) with oral anticoagulants may lead to slight variations of INR values. In this case, increased monitoring of INR values should be conducted during the period of concomitant use as well as for 1 week after paracetamol treatment has been discontinued.

PREGNANCY AND LACTATION
Pregnancy:
Clinical experience of intravenous administration of paracetamol is limited. However, epidemiological data from the use of oral therapeutic doses of paracetamol indicate no undesirable effects on the pregnancy or on the health of the foetus / newborn infant.
Prospective data on pregnancies exposed to overdoses did not show an increase in malformation risk.
Reproductive studies with the intravenous form of paracetamol have not been performed in animals. However, studies with the oral route did not show any malformation of foetotoxic effects.
Nevertheless, IFIMOL IV should only be used during pregnancy after a careful benefitrisk assessment. In this case, the recommended posology and duration must be strictly observed.
Lactation:
After oral administration, paracetamol is excreted into breast milk in small quantities. No undesirable effects on nursing infants have been reported.
Consequently, IFIMOL IV may be used in breast-feeding women.

SIDE EFFECTS
As all paracetamol products, adverse drug reactions are rare (>1/10000, <1/1000) or very rare (<1/10000), they are described below:

Organ system	Rare >1/10000, <1/1000	Very rare <1/10000
General	Malaise	Hypersensitivity reaction
Cardiovascular	Hypotension	
Liver	Increased levels of hepatic transaminases	
Platelet/blood		Thrombocytopenia, Leucopenia, Neutropenia.

Frequent adverse reactions at injection site have been reported during clinical trials (pain and burning sensation).
Very rare cases of hypersensitivity reactions ranging from simple skin rash or urticaria to anaphylactic shock have been reported and require discontinuation of treatment.
Cases of erythema, flushing, pruritus and tachycardia have been reported.
Cutaneous hypersensitivity reactions including skin rashes, angioedema, Stevens Johnson
Syndrome/Toxic Epidermal Necrolysis have been reported.

SYMPTOMS AND TREATMENT OF OVERDOSE
There is a risk of liver injury (including fulminant hepatitis, hepatic failure,cholestatic hepatitis, cytolytic hepatitis), particularly in elderly subjects, in young children, in patients with liver disease, in cases of chronic alcoholism, in patients with chronic malnutrition and in patients receiving enzyme inducers. Overdosing may be fatal in these cases.
- Symptoms generally appear within the first 24 hours and comprise: nausea, vomiting, anorexia, pallor, abdominal pain. Overdose, 7.5 g or more of paracetamol in a single administration in adults and 140 mg/kg of body weight in a single administration in children, causes hepatic cytolysis likely to induce complete and irreversible necrosis, resulting in hepatocellular insufficiency, metabolic acidosis and encephalopathy which may lead to coma and death.

Simultaneously, increased levels of hepatic transaminases (AST, ALT), lactate dehydrogenase and bilirubin are observed together with decreased prothrombin levels that may appear 12 to 48 hours after administration.
Clinical symptoms of liver damage are usually evident initially after two days, and reach a maximum after 4 to 6 days.
Emergency measures
- Immediate hospitalisation.
- Before beginning treatment, take a tube of blood for plasma paracetamol assay, as soon as possible after the overdose.
- The treatment includes administration of the antidote, N-acetylcysteine (NAC), by the i.v. or oral route, if possible before the 10th hour. NAC can, however, give some degree protection even after 10 hours, but in these cases prolonged treatment is given.
- Symptomatic treatment.
- Hepatic tests must be carried out at the beginning of treatment and repeated every 24 hours. In most cases hepatic transaminases return to normal in one to two weeks with full restitution of liver function. In very severe cases, however, liver transplantation may be necessary.

STORAGE
Store below 30°C.

PRESENTATION
100 mL FFS bottle.

SHELF LIFE
24 months

Manufactured by:
UNIQUE PHARMACEUTICAL LABORATORIES
(A Div. of J. B. Chemicals & Pharmaceuticals Limited)
Plot No.4, Phase IV, GIDC Industrial Area,
Panoli 394 116, Gujarat State, India
Revised date: May 2013

Date of Revision - August 2023

137313

440 MM

140 mm

SAP Code: 137313	Country : Malaysia	Mfg. Location : Panoli IV-17	Product Name : IFIMOL IV
Packaging Material : Leaflet	Existing / Reference Art : 127590	Version No.: 1	Date : 24.08.2023
Dimension : (L) 440 mm x (B) 140 (± 1mm tolarance)		Size After Folding : 140 x 27.5 ± 1 mm	Core Dia : NA
Reel Dia : NA	Varnish / Lamination : NA	Roll Unwind Direction : NA	Print Repeat Length : NA
Specification : ITC fine paper		Pantone : Black	
Grammage: 40 ± 5 gm/m^2		Nature of Folds : 4 Horizontal folds.	
Thickness : NA			
Other Test : NA			
Reason for Change : Change in text of Method of Administration			J. B. CHEMICALS & PHARMACEUTICALS LTD.
File Path : X:\Artworks\Prasad\Malaysia\IFIMOL IV 100ml V-1\FFS\open file\137313 IFIMOL IV Leaflet - Malaysia.cdr			

Artist : Shrikant

This document is equivalent to a digitally certified copy & digital signature is not required.